

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

May 2014

EC 309.01 Digital Communication

Date: 08.05.2014, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q-1 (a) The PDF of amplitude x of a certain signal $x(t)$ is given by $p_x(x) = 0.5|x|e^{-|x|}$. [03]

1. Find the probability that $x \geq 1$.
2. Find the probability that $-1 < x \leq 2$.
3. Find the probability that $x \leq -2$.

(b) A Source emits one of four messages randomly every 1 microsecond. The probabilities of these messages are 0.5, 0.3, 0.1, and 0.1. Messages are generated independently. [04]

1. What is the source entropy?
2. Obtain a compact binary code and determine the average length of the code word, the efficiency, and the redundancy of the code.

(C) A television signal has a bandwidth of 4.5 MHz. This signal is sampled, quantized, and binary coded to obtain a PCM signal. [04]

1. Determine the sampling rate if the signal is to be sampled at a rate 20% above the nyquist rate.
2. If the samples are quantized into 1024 levels, determine the number of binary pulses required to encode each sample.
3. Determine the binary pulse rate (bits per second) of the binary - coded signal, and the minimum bandwidth required to transmit this signal.

Q-2 Attempt any two [12]

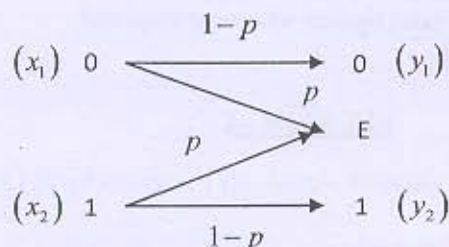
(a) The joint PDF $p_{xy}(x, y)$ of two continuous RVs is given by [06]

$$p_{xy}(x, y) = xye^{-(x^2+y^2)/2} u(x)u(y)$$

1. Find $p_x(x)$, $p_y(y)$, $p_{x/y}(x/y)$, and $p_{y/x}(y/x)$.
2. Are x and y independent?

- (b) In data communication using error detection code, as soon as an error is detected, an automatic request for retransmission (ARQ) enables retransmission of the data in error. In such a channel the data in error is erased. Hence, there is an erase probability p , but the probability of error is zero. Such a channel, known as a binary erasure channel (BEC), can be modeled as shown in figure. [06]

Determine $H(x)$, $H(x/y)$, and $I(x, y)$ assuming the two transmitted messages equiprobable.



- (c) Shows that the output SNR increases exponentially with the bandwidth expansion factor B_T / B . [06]
- (d) Explain briefly detection error probability. [06]

Q-3 Attempt any two [12]

- (a) Shows that, [06]

1. $\log M \geq H(x) \geq 0$, where $H(x)$ is entropy of source x and M is number of different messages.
2. $H(x) = 0$, if all probabilities are zero, except for one, which must be unity.
3. $H(x) = \log M$, if all the probabilities are equal so that $p(x)_i = p_i = \frac{1}{M}$ for all i 's.

- (b) Do as Direct [06]

1. In a random experiment, a trial consists of four successive tosses of a coin. If we define an RV x as the number of heads appearing in a trial, determine $P_x(x)$ and $F_x(x)$. [04]
2. Define Following Terms [02]
 1. Automatic Repeat Request
 2. Forward Acting Error Correction

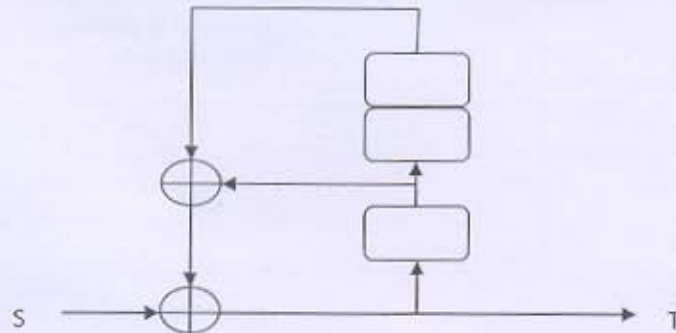
- (c) Show that S_0/N_0 , the SNR in PCM, is a linear function of the message signal power [06]

$\overline{m^2(t)}$ for $\mu = 0$ with the help of necessary derivation and figure.

SECTION - II

Q-4 (a) Data at a rate of 6 kbit/s is to be transmitted over a leased line of bandwidth 4 kHz [02]
using nyquist criterion pulses. Determine the maximum value of the roll off factor r
that can be used.

(b) A scrambler is shown in figure. If the sequence $S = 101010100000111$ is applied to [04]
the input of this scrambler determine the output sequence T .



(c) For a PCM signal, determine L if the compression parameter $\mu = 100$ and the [03]
minimum SNR required is 45 dB. Determine the output SNR with this value of L .
Remember that L must be a power of 2, that is, $L = 2^n$ for a binary PCM.

(d) Three regular dice are thrown. Assign probabilities to the following events: the sum [02]
of the points appearing on the three dice is: (a) 4; (b) 9; (c) 15.

Q-5 Attempt any two [12]

(a) What are the desirable properties of line coding? Also give comparative study of [06]
different line codes.

(b) Explain the ISI and signal degradation study using eye diagram. [06]

(c) What is the slope overload distortion and granular noise in delta modulation and how [06]
it is removed in Adaptive delta modulation?

Q-6 Attempt any three [12]

(a) In binary data transmission using duo binary pulses, sample values of the received [04]
pulses were read as follows: (First bit in the sequence is one)

1 2 0 0 0 -2 0 0 -2 0 2 0 0 -2 0 2 2 0 -2

1. Explain if there is any error in detection.
2. Can you guess the correct transmitted digit sequence? There is more than one possible correct sequence. Give as many as possible correct sequences.

assuming that more than one detection error is extremely unlikely.

- (b) Explain the generation of coherent binary frequency shift keying (BFSK) modulation technique. Draw the constellation diagram for same. [04]
- (c) Compare digital carrier systems ASK, FSK, PSK. [04]
- (d) Write a short note on Central limit theorem. [04]
